

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the principles of vaccination	To include the different forms of vaccines (live vaccine, dead microorganisms, pathogen fragments).
(b) the programme of vaccination used in the United Kingdom	To include a discussion of the reasons for changes to the programme and the importance of boosters. HSW7
(c) the role of vaccination programmes in the prevention of epidemics	To include reference to the establishment of herd immunity.
(d) the biological problems in the development of vaccines and the use of vaccination programmes	To include issues with vaccine development, mutation rate and antigen variability (e.g. in HIV and the influenza virus(es)) and live vaccines AND vaccine use – storage of vaccine, distribution of vaccine and the nutritional status of the target population e.g. if protein deficient.
(e) the ethical issues related to the development and use of vaccines	To include the use of a vaccine in girls against Human Papilloma Virus (HPV) to prevent cervical cancer. HSW10
(f) the use of antibiotics in the treatment of communicable disease	To include an outline of the modes of action of antibiotics e.g. inhibition of bacterial protein, DNA and cell wall synthesis AND the cellular differences between prokaryotic and eukaryotic cells that allow antibiotics to act on bacterial but not human cells.
(g) how the misuse of antibiotics can lead to the evolution of resistant strains of bacteria	To include reference to TB and MRSA.
(h) (i) the precautions taken to reduce the spread of resistant bacteria in hospitals	
(ii) practical investigation of the effect of antibiotics on Gram-positive and Gram-negative bacteria.	To include bacteriostatic and bacteriocidal effects, best hygiene practices, and isolation procedures. PAG1, PAG7 HSW3, HSW4, HSW5, HSW6, HSW8